Moving the sun in architecture photos

Mehmet Ozturk | github.com/mehmetztrk/sun-direction-relighting

What this project is about

Architects often want to see the same building at a different time of day, or with the sun on another side. Taking a new photo or setting up a new render takes time, so it would be great to change the sun with AI in a few seconds.

The hard part is that the sun changes a lot of things at once. Shadows have to move, some walls get brighter, others get darker, and reflections change. But the building itself must not change at all. Windows, materials and furniture have to stay the same. AI models like to redraw things, and that is exactly what an architect doesn't want.

So my goal was simple: find out which tools can do this, try them on real photos, show what works and what breaks, and say what I would use in a real product.

What I did

- I looked at what's out there: open models, LoRAs, paid APIs and apps.
- I picked three real photos from Unsplash: a villa, a concrete apartment block and a living room.
- I tried three models on all three photos: Sun-Direction, Qwen 3D Lighting and FLUX.1 Kontext.
- I used free Hugging Face demos first. When the free quota ran out, I ran Sun-Direction on my own RTX 4070.
- The models kept redrawing things, so I wrote my own script that copies only the light onto the original photo.
- I tested the villa with the sun in many positions, to see if the direction is really visible.

How I judged the results

I asked three simple questions: did the light really change, did the building stay the same, and did single objects like the sofa stay the same? I also used a small script that scores each result, but it missed some mistakes, so the final decision was always by eye.

Results in one look

- **Sun-Direction:** the best one. It really moves the sun on exteriors, but it changes some materials.
- **Qwen 3D Lighting:** fast and keeps the colors, but it zooms the photo and old shadows stay.
- **FLUX.1 Kontext:** keeps the building, but the sun doesn't move on exteriors.
- **My light transfer script:** furniture stays perfect, but the light barely changes.
- **Interiors:** nothing worked well enough yet.

Time and cost

- Sun-Direction: 20 to 26 s on a free Hugging Face Space, 73 to 116 s on my RTX 4070
- Qwen 3D Lighting: 6.5 to 24 s on a free Space
- FLUX.1 Kontext: 31 to 45 s on a free Space
- Everything cost me $0. Space times include the queue.

What I would ship

The Sun-Direction 4B model on a serverless GPU, for exteriors only, with an automatic check that retries if the building changes. My estimate is about $0.01 to $0.02 per image and 20 to 26 s. That estimate uses RunPod prices and my Space times. I did not test it on that GPU.

How I measured

Every number here comes from my compare.py script. It helps me sort results, but it doesn't replace looking at them.

- **Structure (0 to 1):** local correlation of brightness in a 15 x 15 pixel window, after undoing zoom and shift. Flat areas like sky are skipped. New shadows lower it too: fake hard shadows scored 0.92 to 0.96, so a low score isn't always damage.
- **Light change (EV):** how much the brightness pattern changed across the image, with the overall exposure change removed.
- **Scale:** the zoom needed to line the result up with the original. 1.00 means the framing held.
- **What it can't do:** it doesn't check if a shadow points the right way. That part was by eye.

The three test photos



Villa
Glass, a pool and a steel sculpture. Lots of ways to go wrong.



Concrete block
A strong diagonal shadow, so you can see if the sun really moved.



Living room
Light can only come through the windows, and there is a lot of furniture.

1. Sun-Direction (9B version, free demo)

What I did: this model is made just for moving the sun. You give it a picture of a lit ball, and it copies that light. It first makes the photo look overcast, then adds the new sun.



Villa, original



Sun moved to the left
The sun moved, but the colors and the pool went gray.



Concrete block, original



Sun moved to the right
The sun moved, but beige concrete turned gray and the sky went pale.



Living room, original



Low sun from the back left
The sofa, table, rug and the view outside all changed.

Setup

FLUX.2 Klein 9B with the Sun-Direction LoRA, on a free ZeroGPU Space. It runs two passes at about 1 megapixel: an overcast pass without the LoRA, then a LoRA pass that gets the overcast image and the ball as references. 4 steps, guidance 1.0, seed 42.

Result

On both exteriors the lit side of the building really changed. Light change was 0.37 to 0.49 EV and the framing held (scale 0.97 to 1.00). Structure was only 0.33 to 0.52, because the image is generated again rather than edited. Output is about 1264 wide, so detail from the 1536 px input is lost too.

Problems and why they happen

- **Color loss:** concrete and the pool went gray. On the 4B version I saved the overcast image and the color was still there, so the loss happens in the LoRA pass, not the overcast pass.
- **Interiors:** the author says the LoRA has no interior training data. It doesn't know light has to come through the windows, and it redrew the view, sofa, table and rug.
- **License:** the LoRA is Apache-2.0, but the 9B base model is non-commercial, so this version can't ship.

2. Sun-Direction (4B version, on my GPU)

What I did: 4B is the smaller version, and it's the one that allows commercial use. I ran it on my own RTX 4070, once with no prompt and once with a short prompt asking it to keep the colors.



Villa, original



No prompt
A lot of clouds appeared and everything is pale.



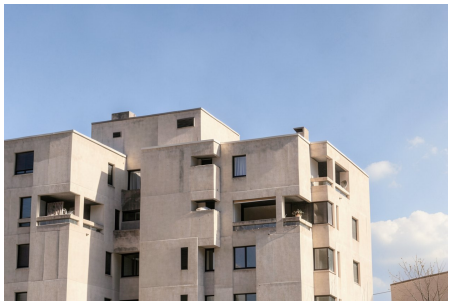
With color prompt
The blue sky came back. The sculpture looks like stone.



Concrete block, original



No prompt
The sun moved, but the concrete faded and clouds appeared.



With color prompt
Blue sky came back, but the concrete is paler and there's a new tree.



Living room, original



No prompt
Nice light, but the view outside is new and the armchair is gray.



With color prompt
Warmer light. Rug and armchair colors changed, and there's an extra ceiling light.

Setup

FLUX.2 Klein 4B (Apache-2.0) with its matching LoRA, on an RTX 4070 with 12 GB. Same two passes, 4 steps and guidance 1.0 through diffusers, with bfloat16 and model CPU offload. Peak GPU memory was 10.4 GB. Color prompt: clear blue sky, warm natural sunlight, same colors and materials as the original.

Result

The color prompt brought back the blue sky and warmer walls without hurting structure much (villa 0.29 to 0.31, block 0.52 to 0.48, room 0.48 to 0.49). It's a cheap fix, and this is the version I would build on, but only for exteriors.

Problems and why they happen

- **Materials:** steel turns to stone because nothing protects it. Every pixel is generated again.
- **Added objects:** the tree and wire on the concrete block are already there in the overcast pass. That pass is a plain edit with no LoRA, so the extra content starts before the sun is even moved.
- **Speed:** 73 to 116 s. The model doesn't fully fit in 12 GB, so parts move between RAM and GPU. Time on a bigger GPU is still unknown.

3. Qwen 3D Lighting (free demo)

What I did: you choose the light direction with an angle, and the model edits the photo.



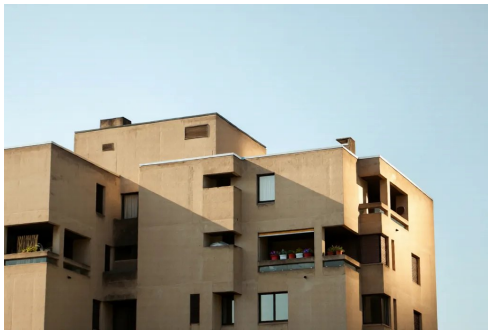
Villa, original



Sun moved to the left
Soft haze on the left, and the photo got zoomed.



Concrete block, original



Sun moved to the right
The right wall got brighter, but the old shadow stayed and flower pots appeared.



Living room, original



Low sun from the back left
Furniture stayed, but the sun patch on the window wall can't happen.

Setup

Qwen-Image-Edit on a free ZeroGPU Space. The Space turns the angle into a prompt such as Light source from the Right. 4 steps, guidance 1.0, output about 1024 wide.

Result

The fastest tool at 6.5 to 24 s, and it keeps colors best. Light change was 0.38 to 0.50 EV, but it's mostly brightness added on top of the old lighting. After undoing the zoom, structure was 0.43 to 0.60.

Problems and why they happen

- **Zoom:** scale came back 0.81 to 0.84 on all four tries, so the output doesn't line up with the original. Using the Space's own image size gave the same result, so it isn't caused by my settings.
- **Light is added, not moved:** old shadows stay. Direction is only a few words in a prompt, and the model doesn't use any depth or geometry. That's probably why it put a sun patch on the window wall.
- **Added objects:** flower pots appeared on the concrete block balcony.

4. FLUX.1 Kontext (free demo)

What I did: this is a normal text edit model. I asked in plain text: change only the lighting and keep the building the same.



Villa, original



Sun moved to the left
Almost nothing changed. Colors are a bit stronger.



Concrete block, original



Sun moved to the right
The diagonal shadow did not move, and the frame shifted a little.



Living room, original



Low sun from the back left
Nice sunset, but everything is orange and the view is new.

Setup

FLUX.1 Kontext [dev] on a free ZeroGPU Space. 28 steps, guidance 2.5, seed 42. It only takes text, so there's no real way to give it a direction.

Result

The best structure on exteriors (villa 0.95, block 0.81), but the weakest light change of the three models on those photos (0.24 and 0.31 EV). In this case a high score mostly means that little happened.

Problems and why they happen

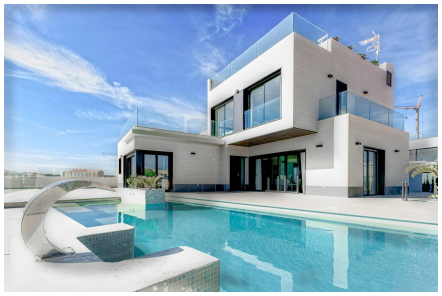
- **No sun movement:** my guess is that keep everything the same beat move the sun in the prompt.
- **Interior:** a strong sunset, about 0.8 EV darker overall, but the sun landed on the wrong side and the outside view was replaced.
- **Frame shift:** the concrete block moved by about 38 px (scale 0.98).
- **License:** the dev weights are non-commercial, so a product would need the paid Pro API, which I did not test.

5. My light transfer script

What I did: the models kept redrawing things, so I kept the original photo and copied only the brightness change from each model's result.



Villa, original



From Sun 9B, smooth
Slight glow at the edges, and the sun didn't move.



From Sun 9B, sharp
Stronger glow along the edges.



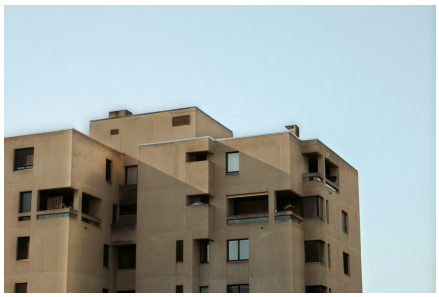
From Qwen
Almost the same as the original.



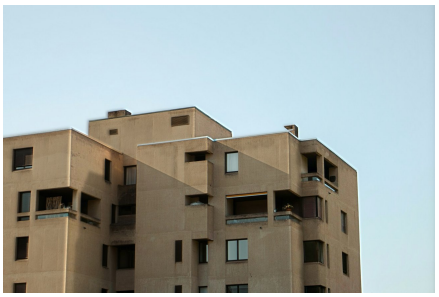
From Kontext
Almost the same as the original.



Concrete block, original



From Sun 9B, smooth
The old shadow is still there.



From Sun 9B, sharp
The old shadow stayed, with a bright halo on the roof.



From Qwen
A small change on the right side.



From Kontext
The shadow got a bit darker.



Living room, original



From Sun 9B, smooth
Furniture is safe, and the light is soft.



From Sun 9B, sharp
More light, but halos and a ghost ring on the floor.



From Qwen
A small bright area on the left.



From Kontext
The room got a bit darker.

Setup

Align the result to the original, take the log2 brightness ratio, smooth it with a guided filter that uses the original as its guide, clip to 2 EV, scale it up and multiply in linear light. Smooth: radius 12, epsilon 0.002. Sharp: radius 3, epsilon 0.05. It runs on a CPU and keeps the full resolution.

Result

Objects are safe: structure 0.94 to 1.00, with no redrawn furniture. But light change falls to 0.04 to 0.20 EV, against 0.24 to 0.61 for the models, so most of the new light is lost on the way.

Problems and why they happen

- **Old shadows stay:** erasing a hard shadow needs a sharp ratio exactly on the shadow edge. Smoothing blurs it, and the model's result doesn't line up well enough for a sharp ratio.
- **Halos:** with less smoothing, edges that don't match between the two images show up as bright lines.
- **Ghosts:** mistakes in the model's result come along too, like the ring shadow on the living room floor.

6. Extra test: the villa with the sun in different spots

What I did: my first villa results didn't show the sun direction clearly. So I gave Sun-Direction 4B a stronger sun and tried nine positions. I wanted to do the same with Qwen and Kontext, but the free quota ran out after one Qwen image, so this test is mostly one model.



Original



Sun behind the camera
Even light on the front.



Sun behind the building
The sun shows in the sky and the walls go into shade.



Sun straight above
Short, hard shadows under the overhangs.



Higher sun from the left
Soft. Too close to the right side version.



Higher sun from the right
So I lowered the sun for the next two.



Low sun from the left
Long shadows across the terrace.



Low sun from the right
The right wall lights up. Left and right are still the closest pair.



Sunset on the left
The biggest change of all.



Sunset on the right
A miss: I asked for the right, but the orange sky stayed on the left.



Qwen, sun behind the camera
My one extra Qwen try. Colors are nice, but it's zoomed in a lot.

Setup

Same 4B setup. I raised the ball brightness from 1.5 to 3.0, set sun height from 8 to 85 degrees, and added strong direct sunlight, hard sharp shadows to the prompt. The sunset shots used their own prompt.

Result

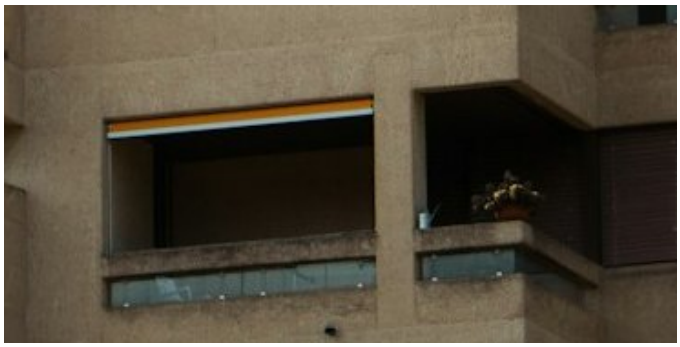
This is the best result of the project. You can tell where the sun is, especially from behind the building, from above and at sunset. Light change was 0.22 to 0.47 EV in daylight and 0.69 to 0.80 EV at sunset.

Problems and why they happen

- **Materials:** the sculpture looks like stone, the pool loses its turquoise and tile lines get softer. Structure dropped to 0.19 to 0.38 here, so stronger light costs more detail.
- **Left and right:** this is the closest pair. The camera sees two faces of the villa at an angle, so the difference is small.
- **Sunset direction:** the right side sunset stayed orange on the left. My guess is that the word sunset pulled the result toward one look, and the ball lost that fight.

Failed results up close

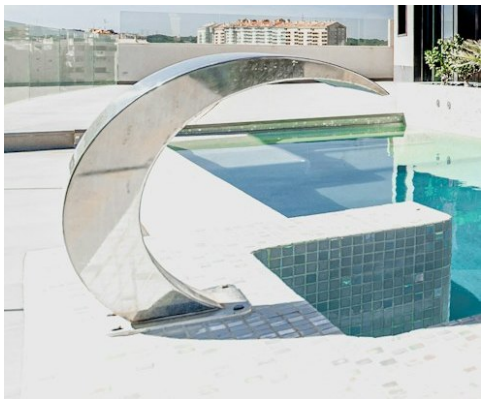
These are the mistakes I want to be clear about. An architect would notice all of them.



Original



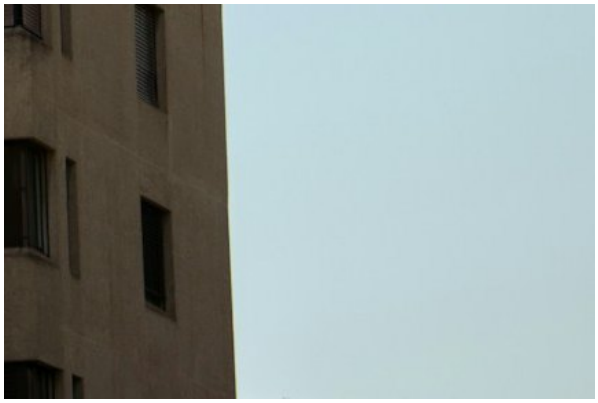
Qwen added flower pots
There were no pots here.



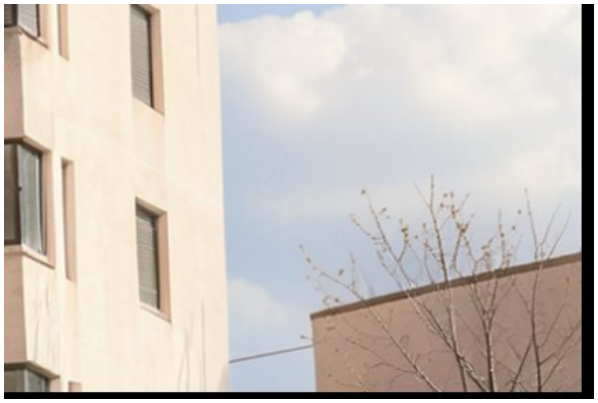
Original



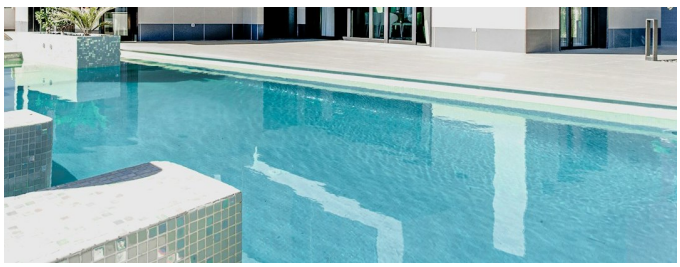
Sun 4B changed the sculpture
Shiny steel now looks like stone.



Original



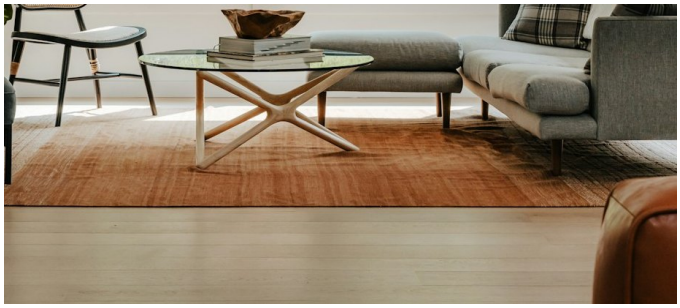
Sun 4B added a tree
This came from the overcast step.



Original



Sun 9B turned the pool gray
The turquoise water is gone.



Original



Sun 9B redrew the table and rug
Note the strange ring shadow on the floor.



Original



Qwen's impossible sun patch
Sun can't hit the wall that holds the windows.



Original



Kontext replaced the view
A new outdoor scene, and the sun is on the wrong side.



Original



Sun 4B also replaced the view
The hillside became trees and sky.



Original



Kontext didn't move the shadow
I asked for the sun on the right. Nothing moved.



Original



My script left a halo
See the bright line along the roof.



Asked for sunset on the left
This one worked.



Asked for sunset on the right
The orange sky stayed on the left.

Furniture check

I compared each object up close, and this showed what the overall picture hid. Black areas fell outside that model's frame.

Armchair

Sun 9B and 4B lost the green color.



Original



Sun 9B



Sun 4B



Qwen



Kontext



Light transfer

Sofa

Sun 9B redrew it.



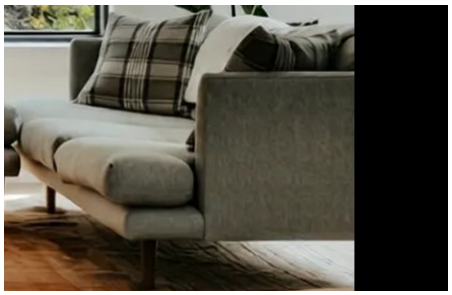
Original



Sun 9B



Sun 4B



Qwen



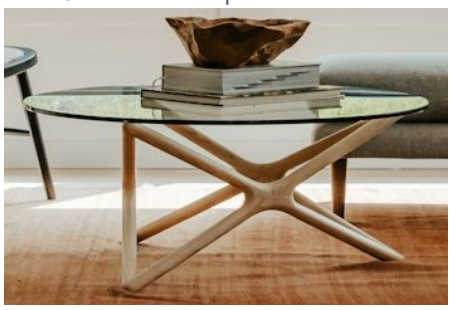
Kontext



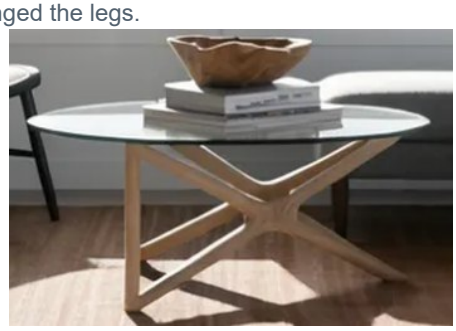
Light transfer

Coffee table

Sun 9B made the top look white and changed the legs.



Original



Sun 9B



Sun 4B



Qwen



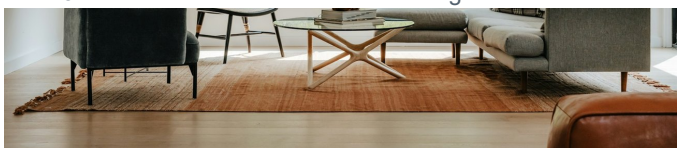
Kontext



Light transfer

Rug

Sun 9B and 4B made it smaller and changed its color.



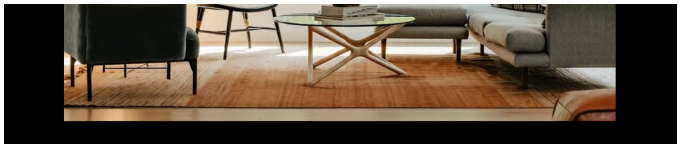
Original



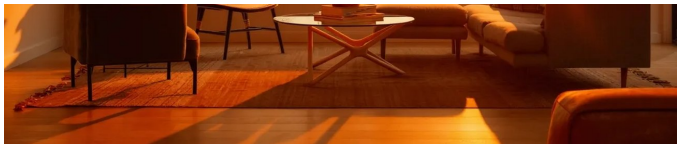
Sun 9B



Sun 4B



Qwen



Kontext



Light transfer

All the numbers

Each cell shows structure / light change in EV / scale. One seed per try, so treat these as a rough guide. They're also in results/_report/scores.csv on GitHub.

Tool	Villa	Concrete block	Living room
Sun 9B	0.33 / 0.37 / 0.97	0.52 / 0.49 / 1.00	0.40 / 0.40 / 0.99
Sun 4B, no prompt	0.29 / 0.30 / 0.97	0.52 / 0.50 / 0.99	0.48 / 0.29 / 0.99
Sun 4B, color prompt	0.31 / 0.27 / 0.97	0.48 / 0.61 / 0.99	0.49 / 0.31 / 0.99
Qwen 3D Lighting	0.43 / 0.42 / 0.81	0.60 / 0.41 / 0.84	0.60 / 0.50 / 0.82
FLUX.1 Kontext	0.95 / 0.24 / 1.00	0.81 / 0.30 / 0.98	0.65 / 0.37 / 1.00
Transfer from Sun 9B, smooth	0.97 / 0.14 / 1.00	0.98 / 0.20 / 1.00	0.96 / 0.15 / 1.00
Transfer from Sun 9B, sharp	0.95 / 0.15 / 1.00	0.94 / 0.20 / 1.00	0.94 / 0.15 / 1.00
Transfer from Qwen	0.99 / 0.07 / 1.00	0.99 / 0.09 / 1.00	1.00 / 0.04 / 1.00
Transfer from Kontext	1.00 / 0.10 / 1.00	1.00 / 0.04 / 1.00	0.98 / 0.16 / 1.00

Villa sun positions

Sun position	Villa, Sun 4B
Behind the camera	0.30 / 0.31 / 0.98
Behind the building	0.24 / 0.47 / 0.98
Straight above	0.38 / 0.22 / 0.98
Higher left	0.32 / 0.27 / 0.98
Higher right	0.19 / 0.39 / 0.98
Low left	0.27 / 0.31 / 0.98
Low right	0.19 / 0.40 / 0.98
Sunset left	0.26 / 0.69 / 0.97
Sunset right	0.27 / 0.80 / 0.97

Time per image

- Sun 9B on a free Space: 20.5 to 26 s, queue included
- Sun 4B on my RTX 4070: 73 to 116 s
- Qwen on a free Space: 6.5 to 24 s, queue included
- Kontext on a free Space: 31.5 to 45 s, queue included

How I would build it

- **Model:** Sun-Direction 4B with a color prompt, because its base model and LoRA are both Apache-2.0.
- **Hosting:** a serverless GPU that only runs when someone uses it. An always-on RTX 4090 costs about $650 a month, which only beats a $0.04 API above about 16,000 images a month.
- **Cost:** RunPod serverless RTX 4090 is $1.34 an hour. At about 25 s per image that's about $0.01, or $0.02 with one retry.
- **Quality gate:** after generating, compare the result with the original and retry or reject it if the structure drops too far. I haven't tuned that threshold yet.
- **Risks:** I didn't test 4B on that GPU or measure cold starts, so the speed and cost above are estimates from 9B Space times.

Problems along the way

- Some free Hugging Face demos were paused or crashing, so I found others.
- The free GPU quota ran out quickly, so I moved the Sun-Direction tests to my own computer.
- The bigger 9B model doesn't fit on my 12 GB card, so I used 4B.
- My saved Hugging Face token was invalid, so runs kept failing after I logged in. Logging in again through the browser fixed it.
- My first score was misleading. It fell near zero when a model zoomed, so I added alignment. Kontext still scored best because it barely changed anything, so I also check by eye.
- I thought the gray colors came from the overcast step. They came from the next step, so a color prompt helped.
- Sun-Direction needed 150 s of quota when I had 147 s left. That's what pushed me to run it locally.

What can be improved

- Test on at least 30 exteriors, since three photos isn't many.
- Protect glass, metal and water with masks, so they don't change.
- Use a depth map to work out real shadows from the sun position.
- Combine the model's new shadows with the original textures.
- Use masks or a room-trained model to lock furniture in interiors.
- Test on a bigger GPU to get closer to your 13.4 s average.
- Let the user drag the sun, or pick a date, time and location.
- Test on renders too, since I only used real photos.

Other models worth trying

- **Sun-Direction v2:** the author says it's in progress, with no overcast step and control over light hardness and color.
- **Qwen Light Migration:** copies the light from a reference photo, so you could use a real sunset photo.
- **Qwen Edit 2511 with LightingRemap:** you paint color blocks at the edge of the image to set the light. Still an alpha.
- **Nano Banana (Gemini):** paid prompt-based editing from about $0.03 per image.
- **FLUX.2 Edit and Seedream Edit:** newer paid edit models on fal.
- **GPT Image 2:** OpenAI's edit model. I couldn't confirm its price.
- **LTX-2.3 Relight:** sets sun direction for exterior video, so it could also suit a time-of-day feature.
- **HELIOS:** a research project with sun angle control from GPS. I couldn't confirm that its code is public.

Code and all tries, including the failed ones: https://github.com/mehmetztrk/sun-direction-relighting

Photos from Unsplash: framesforyourheart, Clay Banks, Valentin Gerardi.